

Submission Summary

Conference NameVI International Congress of Interdisciplinary Research #LoHacemosxTIC

Paper ID12

Paper TitleComparison of RLQR and MPC Performance Under Packet Loss Modeled by a Finite-State Markov Channel

Abstract

Recent technological advances have introduced systems requiring closed-loop feedback via available communication systems, especially those implemented in wireless networks. Automated medical assistance, autonomous ground and aerial vehicles, home automation, cooperative robotics, and smart grids, among others, depend on control systems operating over wireless networks. However, wireless communications are susceptible to faults such as delays, external interference, packet loss, and transmission capacity limitations. Among these problems, packet loss can particularly affect the performance and stability of control systems with wireless feedback loops. Therefore, this research presents two optimal control methods to compensate for the faults caused by packet loss in the communication channel. We compare the performance of two controllers to counteract these losses: the Robust Linear Quadratic Regulator (RLQR) and Model Predictive Control (MPC), both implemented on a finite-state Markov channel model to account for the stochastic nature of packet loss. The RLQR is considered because packet loss is treated as a source of uncertainty in the model, under the assumption that the robust regulator can minimize its impact. On the other hand, the MPC is used for its predictive capability, which allows the transmission of a set of control actions per packet, helping prevent performance degradation due to packet loss. Simulation results show the effect of packet loss on the state variables and control actions of the closed-loop system. We quantify the performance by measuring the RMS values of these signals. The results indicate that RLQR achieves acceptable reference tracking at the cost of increased control actions, suggesting sensitivity to packet loss. On the other hand, MPC demonstrates superior performance with a better balance between system throughput and control effort.

Created2026-05-12 22:48:50 GMT-5

Last Modified2026-05-18 21:01:52 GMT-5

Authors**Ricardo Alayon** (Universidad Distrital Francisco José de Caldas)

<raalayonm@udistrital.edu.co>

Andres Jutinico (Universidad Distrital Francisco José de Caldas)

<aljutinicoa@udistrital.edu.co>

Roberto Ferro (Universidad Distrital Francisco José de Caldas) <rferro@udistrital.edu.co>

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Technology and artificial intelligence

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Andrés Leonardo Jutinico Alarcón: PhD in Mechanical Engineering, Professor, +57 3127977123.

Ricardo Arturo Alayon Mora: Bachelor in mathematics. Student of magister in information and communication sciences. +57 3187925853.

Roberto Ferro Escobar: Ph.D in Informatic Engineering, professor. 3138135048

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